

LESSON 2-4 • SESSION 1 • GUIDED NOTES**Building a Box Plot**

Learning target – I can find the five-number summary of a data set and use it to build a box plot. · 6.SP.B.4

WORDS TO KNOW*fill in the missing words*

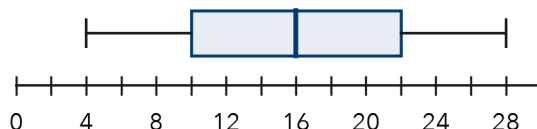
WORD	WHAT IT MEANS	EXAMPLE
five-number summary <i>resumen de cinco números</i>	Min, Q1, _____, Q3 and max.	5, 9, 14, 20, 23
quartile <i>cuartil</i>	Q1 is the median of the _____ half; Q3 is the median of the upper half.	lower half 5, 9, 11: Q1 = 9
box plot <i>diagrama de caja</i>	A graph of the five-number summary: a box from Q1 to Q3 and two _____.	box 9 to 20, line at 14

KEY IDEA*complete each sentence*

- With an odd number of values, leave the _____ out of both halves.
- Each of the four sections of a box plot holds about one _____ of the data.

WORKED EXAMPLE*fill in the blanks as we solve*

Times (min): 14, 5, 20, 9, 23, 11, 16. Find the five-number summary.

Step 1 Order: 5, 9, 11, 14, 16, 20, 23. Median = ____.**Step 2** Lower half 5, 9, 11: Q1 = _____. Upper half 16, 20, 23: Q3 = ____.**Answer** **5, 9, 14, 20, _____****YOU TRY***use the example as a model***1.** Data: 3, 8, 5, 12, 10, 7, 15. Find each value.**a.** Q1: ____**b.** Median: ____**c.** Q3: ____**2.** Read the box plot.**Q1; median; Q3** _____**SUM IT UP***in your own words*

To build a box plot, I first _____, then _____.

LESSON 2-4 • SESSION 2 • GUIDED NOTES

Comparing Box Plots

Learning target – I can compare two groups by reading the medians and box widths of their box plots. • 6.SP.B.4

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
center <i>centro</i>	A typical value; on a box plot, the _____ line.	median 20 min
box width <i>ancho de la caja</i>	Q3 _____ Q1; the spread of the middle half.	$22 - 18 = 4$
consistent <i>constante</i>	Values are _____ together; the box is narrow.	box 18 to 22

KEY IDEA*complete each sentence*

- Compare _____ to see which group has the higher typical value.
- Compare box widths: the _____ box is more spread out.

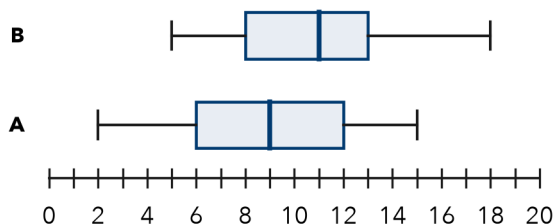
WORKED EXAMPLE*fill in the blanks as we solve*

Trail X: Q1 18, median 20, Q3 22. Trail Y: Q1 19, median 26, Q3 33 (minutes).

Step 1 Medians: $26 - 20 = \underline{\hspace{1cm}}$, so Y is higher by 6 min.**Step 2** Box widths: X is _____, Y is _____.**Answer** Trail _____ is more consistent.**YOU TRY***use the example as a model*

1. Which plot has the higher median? Which is more spread out?

Median; spread _____



2. Find each box width.

a. Q1 = 12, Q3 = 30: _____

b. Q1 = 41, Q3 = 47: _____

SUM IT UP*in your own words*

To compare two box plots, I look at _____ and _____.
